

# Tutorial 3. Interest Points, Registration and Fitting

Chenrui Fan  
Computer Vision 2026  
University of Bern

**Interactive lab:** [cv-tutorial-03-lab.chenrui-fan.chatgpt.site](http://cv-tutorial-03-lab.chenrui-fan.chatgpt.site)

**Source code:** [github.com/Sosekie/Unibe-CVG-CV-Interactive-Labs](https://github.com/Sosekie/Unibe-CVG-CV-Interactive-Labs)

**Scope.** This tutorial focuses on interest-point detection and description, line fitting, and 2D image registration. Q2 and Q6 extend the lecture with Hessian and Prewitt calculations and are optional. State any additional assumptions you use. The solution sheet repeats each question before its answer.

## Interest points

**Focus:** Harris and Hessian responses; contrast scaling; detector versus descriptor.

Use one-based array coordinates  $(r, c)$ , with  $r$  increasing down the page and  $c$  to the right. The displayed image axes have  $x$  to the right and  $y$  upward. Define

$$I^{(1)} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}, \quad I^{(2)} = \frac{1}{2}I^{(1)}.$$

The two locations are  $p_* = (4, 4)$  and  $p_{**} = (6, 4)$ . At an image boundary use *reflection without repeating the boundary pixel*: for example,  $I(0, c) = I(2, c)$  and  $I(7, c) = I(5, c)$ ; likewise for columns.

**Q1. Harris response.** Approximate the lecture's Gaussian derivative and Gaussian window with the following unnormalised central first differences and a uniform  $3 \times 3$  average:

$$I_x(r, c) = I(r, c + 1) - I(r, c - 1), \quad I_y(r, c) = I(r - 1, c) - I(r + 1, c).$$

At each location form the second-moment matrix  $M = \frac{1}{9} \sum_{q \in \mathcal{N}_{3 \times 3}} \nabla I(q) \nabla I(q)^\top$  and the score  $R = \det M - k(\text{tr } M)^2$  with  $k = 0.05$ . Compute  $M$  and  $R$  at  $p_*$  and  $p_{**}$  in both images. Explain how  $R$  changes if every intensity is multiplied by  $\alpha$ .

**Q2. Hessian response (optional extension).** To remove discretisation ambiguity, use the following *direct second differences of intensity*:

$$\begin{aligned} I_{xx}(r, c) &= I(r, c + 1) - 2I(r, c) + I(r, c - 1), \\ I_{yy}(r, c) &= I(r - 1, c) - 2I(r, c) + I(r + 1, c), \\ I_{xy}(r, c) &= \frac{1}{4} [I(r - 1, c + 1) - I(r - 1, c - 1) \\ &\quad - I(r + 1, c + 1) + I(r + 1, c - 1)]. \end{aligned}$$

For  $H = \begin{bmatrix} I_{xx} & I_{xy} \\ I_{xy} & I_{yy} \end{bmatrix}$ , compute  $\det H$  at both locations in both images. What contrast-scaling law does this score obey?

**Q3. Detector and descriptor.** Explain the different jobs of an interest-point detector and a feature descriptor when matching two images. Why are scale and orientation often estimated or normalised?

## Fitting

**Focus:** ordinary and total least squares; the Prewitt operator as a local plane fit.

**Q4. Ordinary least-squares line.** For observations  $(x_i, y_i)$ ,  $i = 1, \dots, n$ , minimise

$$E(c_1, c_0) = \sum_{i=1}^n (c_1 x_i + c_0 - y_i)^2.$$

Derive the normal equations for  $y = c_1 x + c_0$ . Under what condition is the fitted slope unique?

**Q5. Numerical line fit.** Apply Q4 to  $(0, -7)$ ,  $(2, -1)$  and  $(4, 5)$ . Give the fitted line and its sum of squared vertical residuals.

**Q6. Prewitt from plane fitting (optional extension).** In a  $3 \times 3$  image patch, let local  $u$  increase to the right and local  $v$  upward; the top row therefore has  $v = +1$ . Fit  $z(u, v) = au + bv + c$  by ordinary least squares to the nine intensity values  $z_{uv}$  at  $u, v \in \{-1, 0, 1\}$ . Derive  $a$  and  $b$  and write the corresponding  $3 \times 3$  masks. Specify whether you are using correlation or convolution.

**Q7. Total least-squares line.** Minimise the squared perpendicular residuals

$$E(a, b, d) = \sum_{i=1}^n (ax_i + by_i - d)^2, \quad a^2 + b^2 = 1.$$

Derive the solution using the centred point matrix. Then compute one valid  $(a, b, d)$  for the three points in Q5. How does this objective differ from Q4?

## Registration

**Focus:** affine maps, point correspondences, and homography estimation.

**Q8. Affine transformation.** Given corresponding points  $(x_i, y_i) \leftrightarrow (x'_i, y'_i)$ , write the linear equations for the six parameters of a 2D affine map. State the minimum number of pairs for a unique exact solution and the required geometric condition. What changes when more noisy pairs are available?

**Q9. Affine numerical example.** Estimate the map from  $(0, 0) \mapsto (1, -2)$ ,  $(1, 0) \mapsto (3, -3)$  and  $(0, 1) \mapsto (2, 1)$ . Predict the image of  $(1, 1)$ .

**Q10. Homography.** For homogeneous points  $p_i = (x_i, y_i, 1)^\top$  and  $p'_i = (x'_i, y'_i, 1)^\top$ , a nonsingular homography satisfies  $p'_i \sim H p_i$ . Eliminate the unknown per-point scale to obtain two homogeneous linear equations per pair in the nine entries of  $H$ . Explain how to estimate  $H$  with SVD when correspondences are noisy. State the minimum number of pairs, the nondegeneracy condition, and the scale ambiguity.

**Q11. RANSAC for registration.** Putative point correspondences include outliers. Explain how RANSAC estimates a homography from them and why fitting once to all matches may fail. If the inlier fraction is  $w = 0.5$ , the minimal sample contains four matches, and the desired probability of drawing at least one all-inlier sample is  $p = 0.99$ , how many independent random samples are needed? Use the standard approximation that each selected match is independently an inlier, and assume independent nondegenerate samples.